



3ª Série

Matemática

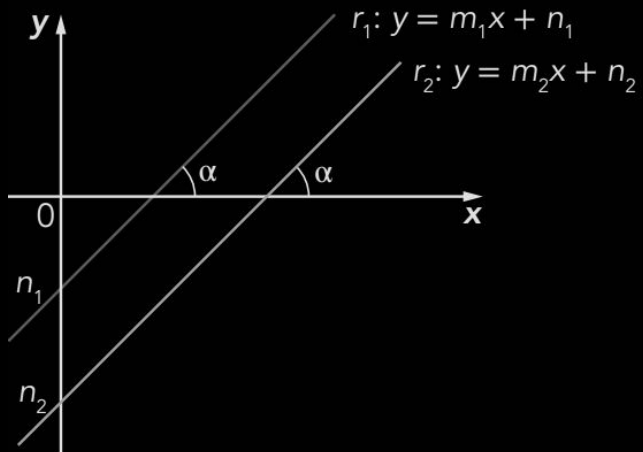
Aula 04 - 2º Bim

Prof. Diogo Pelaes

Paralelismo e Perpendicularidade

- Pg. 252 e 253, Ex. 49 ao 54
- Pg. 255, Ex. 56 ao 61

Paralelismo

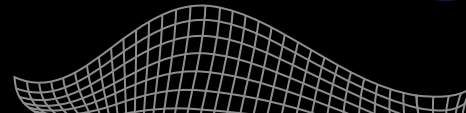
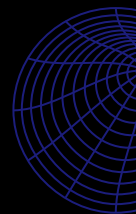
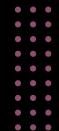
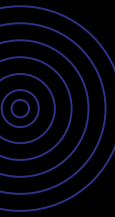


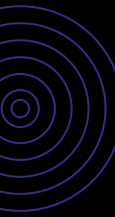
$$r_1 // r_2 \Leftrightarrow m_1 = m_2$$



Exemplo 1

$r: y = 3x - 2$ e $s: 6x - 2y + 5 = 0$

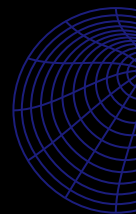




Exemplo 2

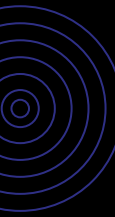
$$r: 3x - y + 7 = 0$$

$$s: 6x - 2y + 14 = 0$$

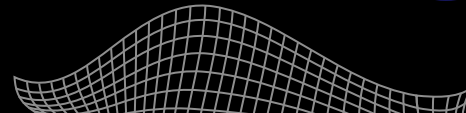
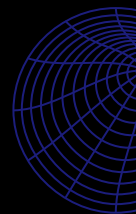
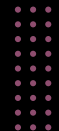




Exercício Resolvido



10. Seja a reta $r: y = 2x - 1$. Obtenha a equação de uma reta s paralela à reta r que passa pelo ponto $P(1, 4)$.

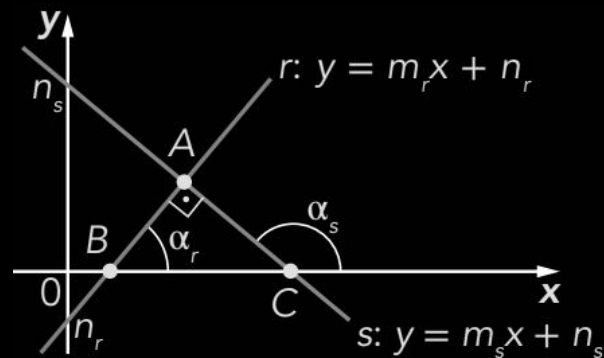


Exercício Resolvido

11. Para que valores reais de k as retas $r: 3x - 2y + 5 = 0$ e $s: kx - y + 1 = 0$ são concorrentes?

Perpendicularidade

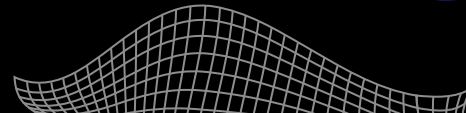
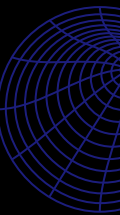
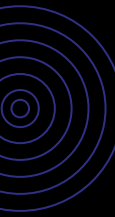
Se r e s são perpendiculares entre si, então $m_r \cdot m_s = -1$.





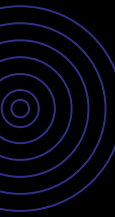
Exemplo 3

$r: 2x - 4y + 5 = 0$ e $s: y = -2x + 3$

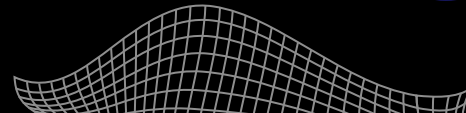
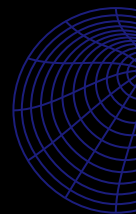
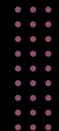




Exercício Resolvido

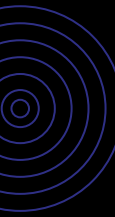


12. Determine a equação geral da reta s , perpendicular a $r: y = 3x + 1$, traçada pelo ponto $P(4, 0)$.

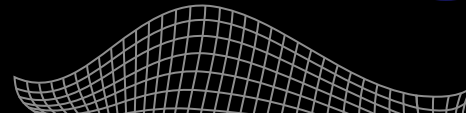
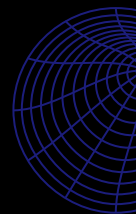
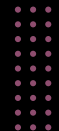




Exercício Resolvido



13. Determine a equação da mediatriz do segmento de reta cujas extremidades são $A(0, 0)$ e $B(2, 3)$.



Exercício Resolvido

14. Determine o ponto de intersecção entre a reta $r: y = \frac{3}{2}x$ e a reta perpendicular a r conduzida pelo ponto $P(-7, 15)$.



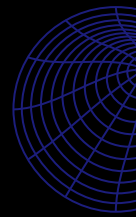
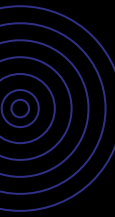
49. Em cada item a seguir, classifique no caderno as retas r e s em paralelas distintas, concorrentes ou coincidentes.

a) $r: y = 4x - 1$ e $s: 8x - 2y + 1 = 0$

b) $r: 5x - y + 6 = 0$ e $s: 6x + y - 5 = 0$

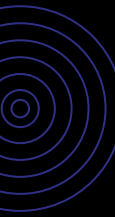
c) $r: y = -\frac{3x}{2} + 2$ e $s: 6x + 4y - 8 = 0$

d) $r: y = -\frac{3x}{4} - \frac{1}{4}$ e $s: 6x + 8y + 4 = 0$





50. Para que valores reais de k as retas de equações $3x + 2y - 1 = 0$ e $kx - 3y + 2 = 0$ são:

- a)** paralelas distintas?
 - b)** concorrentes?
 - c)** coincidentes?
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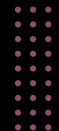
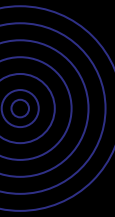


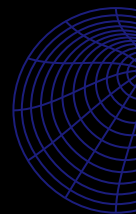
51. Escreva, no caderno, a equação geral da reta s que é paralela a r e passa por P , sendo r em cada um dos casos a seguir.

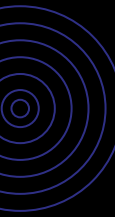
a) $r: y = 3x - 4$ e $P(0, 1)$

b) $r: 2x + 5y - 4 = 0$ e $P(-1, 2)$

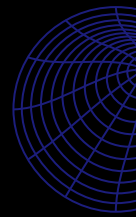
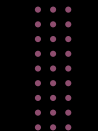
c) $r: y - 3 = 0$ e $P(2, 5)$



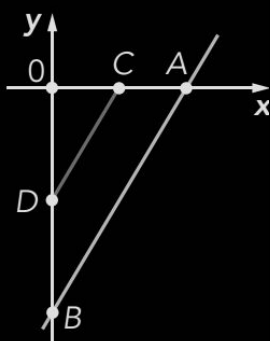


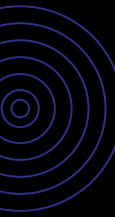
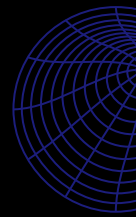


52. Mostre no caderno que o quadrilátero de vértices $A\left(-\frac{4}{5}, \frac{13}{5}\right)$, $B\left(-\frac{1}{2}, \frac{7}{2}\right)$, $C(1, 2)$ e $D\left(\frac{2}{5}, \frac{1}{5}\right)$ é um trapézio.



- 53.** Na figura, a equação da reta que passa por A e B é $5x - 3y - 15 = 0$. Sabendo que C é ponto médio de $\overline{OA}$ e D é ponto médio de $\overline{OB}$, determine no caderno a medida de perímetro do triângulo COD .



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- 54.** As retas suportes de três lados de um paralelogramo são $r: 3x + 2y - 12 = 0$, $s: y = \frac{x}{2} - 1$ e $t: x - 2y + 6 = 0$. Sendo o ponto $(2, 0)$ um dos vértices do paralelogramo, determine no caderno a equação da reta suporte do quarto lado.
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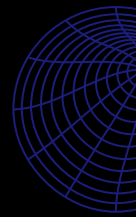
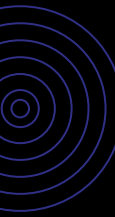
56. Descreva, no caderno, quais das retas a seguir são perpendiculares entre si.

$r: y = 2x + 3$

$s: x - 4y + 4 = 0$

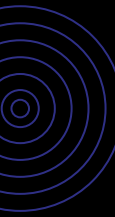
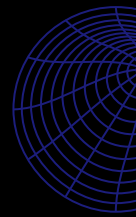
$t: x + 2y - 6 = 0$

$u: y = -2x - 1$



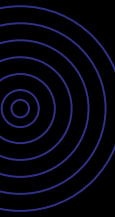
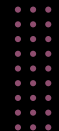


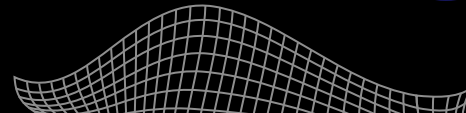
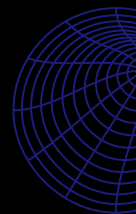
57. Dado o segmento de reta $\overline{AB}$, com $A(4, 5)$ e $B(-2, 1)$:

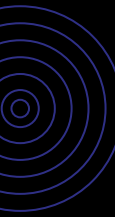
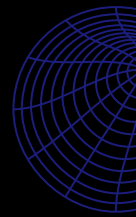
- a)** determine no caderno a equação da mediatriz de $\overline{AB}$;
- b)** escolha um ponto qualquer dessa mediatriz e mostre no caderno que esse ponto equidista de A e B .
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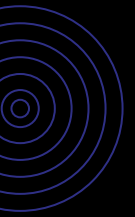


58. Dados $P(2, -4)$ e $r: 2x - 3y + 6 = 0$:

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- a) obtenha no caderno as coordenadas do ponto Q , intersecção de r com a perpendicular a r por P ;
- b) determine no caderno o simétrico de P em relação à reta r .
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- 60.** As retas r e s , de equações $r: 2x - y + 3 = 0$ e $s: y = mx + n$, intersectam-se, perpendicularmente, no ponto $(-2, -1)$. Quais são os valores de m e n ?
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- 61.** Seja ABC o triângulo de vértices $A(0, -3)$, $B(-4, 0)$ e $C(2, 1)$. Determine no caderno a equação da reta suporte da altura relativa ao lado $\overline{BC}$.
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